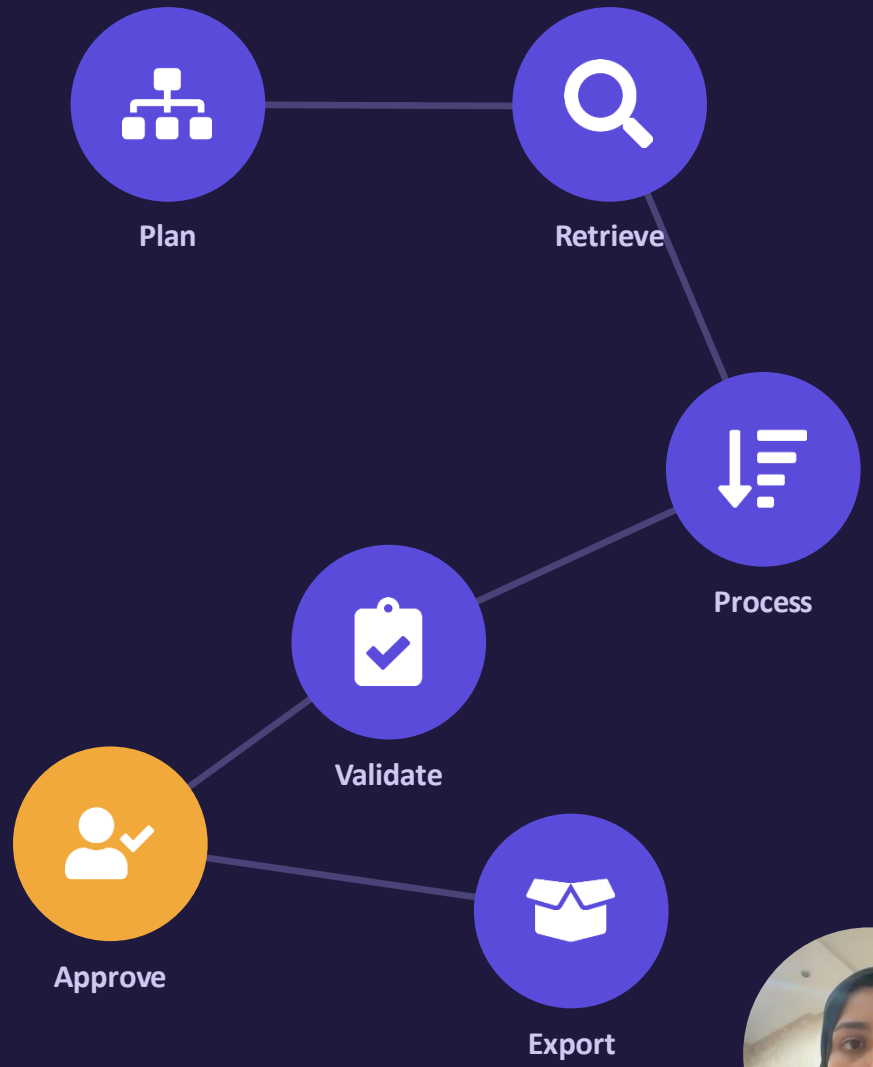
PRACTICAL DEVELOPMENT / LIVE DEMONSTRATION

From research goal to reviewed evidence

An LLM-powered academic research planning agent

Live case: exercise protocols for knee joint pain

Implementation, run evidence, testing and critical review





11 October 2026

One broad question becomes three research tasks

RESEARCH GOAL

What is the best exercise protocol for reducing knee joint pain?

01 Exercise type

Which kinds of exercise have been studied?

02 Dose

How often and for how long were they prescribed?

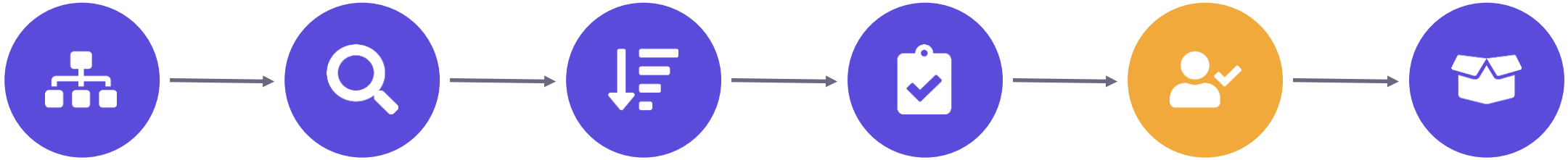
03 Long-term effects

Do pain and knee outcomes persist over time?

Assessment focus

A working agent, documented decisions, test evidence and an honest account of limits.

The workflow is explicit, bounded and reviewable



Plan

Hosted model
2–4 subquestions

Retrieve

Crossref + OpenAlex
live academic records

Process

Deduplicate · rank
Summarise abstracts

Validate

Record IDs · quotes
Independent checks

Approve

Human decision
required

Export

MD · JSON
CSV

Supervisor routes a failed check to one targeted retry; replans once if needed, then stops.

IMPLEMENTATION NOTE

Six functional roles run inside one Python/LangGraph process; they are not six separately deployed agents.

One live run recovered from failed evidence checks

10:34

1

Plan and search

3 subquestions × 2 scholarly APIs per pass

12 searches across two passes

10:34

2

Synthesize

10 unique records · 10 usable abstracts · 6 claims

First evidence package

10:34–10:35

3

Reject and recover

Two quote checks failed; a processing retry also failed

Supervisor requested a replan

10:35

4

Validate and approve

Replanned run passed; user typed APPROVE

Final export completed

Run ID 0e82cee4…b550d7b4

Validation passed · Retrieval/processing errors: 0

The output reports improvement signals, not a winning protocol

TWO SHORT-TERM RESULTS RETURNED BY THE LIVE RUN

Stretching · 8 weeks

37.14 → 28.24

Reported pre/post NRS scores in older adults with genu osteoarthritis.

Swing exercise · 7 days

7.15 → 6.00

Reported mean pain score in one older-adult residential setting.

Why this cannot answer “best”

Different populations, outcome scales and follow-up periods. Abstract-level details are incomplete, so the values cannot be ranked against each other.

Sources: stretching report (2025), doi:10.55927/cjas.v3i8.65 · Domini (2026), doi:10.9790/1959-1502025458

A 377-person trial found no added benefit from higher intensity

START randomized clinical trial · 18-month WOMAC pain outcome

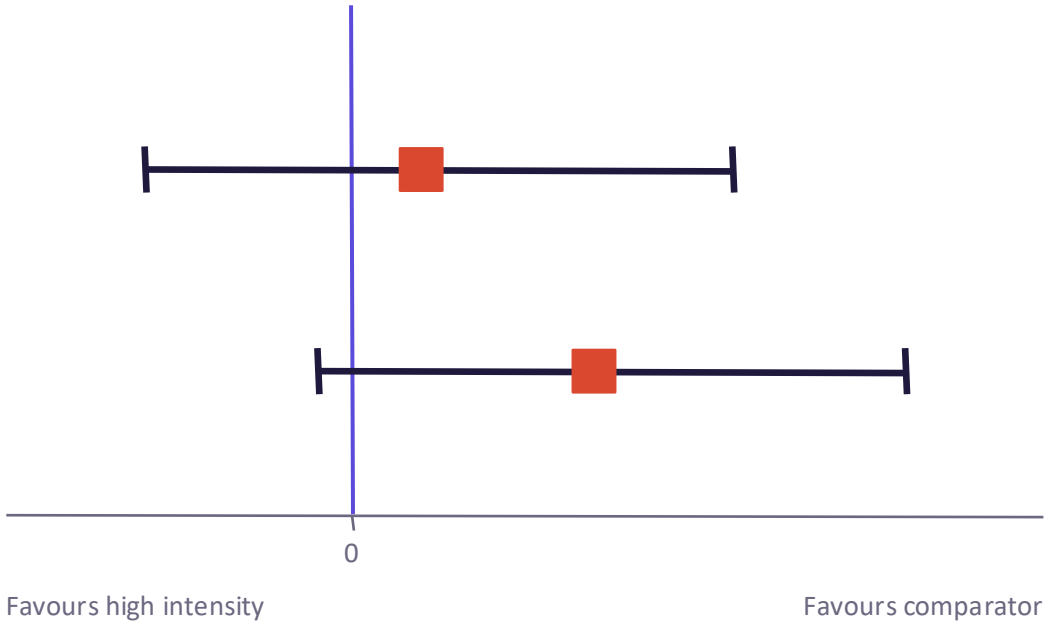
Adjusted difference in pain score (high intensity minus comparator)

High intensity vs attention control

Δ 0.2 [-0.6, 1.1] p = .61

High vs low intensity

Δ 0.7 [-0.1, 1.6] p = .08



Both confidence intervals cross zero: no statistically significant advantage for high intensity.

Messier et al. (2021), JAMA, 325(7), 646–657. doi:10.1001/jama.2021.0411

Export is a human decision, not an automatic side effect

LIVE RUN STATUS

Validated draft

10 records · 10 abstracts
6 source-linked claims



Human review

APPROVE / REJECT

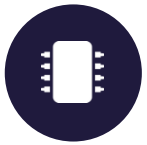
Approved research package

Markdown report · source CSV · run JSON
event log · integrity hashes



Credential handling

The Colab notebook prompts for the Hugging Face token at run time; the token is not written to project files or exports.



Model calls require an enabled provider and available credits.

A validation pass checks attribution and structure; human review must assess whether the evidence supports the conclusion.

Tests show workflow reliability—not research accuracy

55

automated tests passed

Unit and functional checks

96.09%

core statement coverage

Coverage in the tested codebase

10

successful live examples

Evidence package exported after approval

What the tests cover

Routing, structured data, source matching, approval, export, error handling and notebook wiring. Mock clients isolate application behavior from service availability.

What remains unproven

One live run cannot estimate reliability across topics. Abstract matching does not verify every inference. Full-text appraisal and independent human scoring are still required.

The 90% operational target is a future evaluation criterion, not a result claimed here.

Design choices are deliberate—and each has a trade-off

CHOICE	WHY IT FITS	LIMIT / NEXT CHECK
Plan-and-Execute + LangGraph	An inspectable task list and explicit retry routes.	Adds model calls and latency; test on ambiguous goals.
Crossref + OpenAlex	Structured records and resolvable source identifiers.	Metadata/abstract coverage is incomplete; screen full texts.
Lexical relevance score	Simple, transparent and reproducible ranking.	Relevance is not study quality; compare with human ratings.
Single approval gate	A person reviews before consequential export.	Quality depends on reviewer skill and attention.

Next evaluation Run the 24-goal set, compare against direct search, then manually rate relevance and claim support; report the confidence interval and failures.

Design sources: Wooldridge (2009); Padgham & Winikoff (2003); Yao et al. (2023).

References

AGENT DESIGN AND METHOD

Padgham, L. and Winikoff, M. (2003) “Prometheus: a methodology for developing intelligent agents”, in Giunchiglia, F., Odell, J. and Weiß, G. (eds), Agent-Oriented Software Engineering III. Berlin: Springer, pp. 174–185. doi:10.1007/3-540-36540-0_14.

Wooldridge, M. (2009) An Introduction to Multiagent Systems. 2nd edn. Chichester: Wiley.

Yao, S. et al. (2023) “ReAct: synergizing reasoning and acting in language models”. ICLR 2023. Available at: arxiv.org/abs/2210.03629.

LIVE RUN AND SYSTEM SOURCES

Messier, S.P. et al. (2021) “Effect of high-intensity strength training on knee pain and knee joint compressive forces among adults with knee osteoarthritis: the START randomized clinical trial”, JAMA, 325(7), pp. 646–657. doi:10.1001/jama.2021.0411.

Domini, D. (2026) “Effect of swing exercises on knee joint pain among elderly people in selected old age home of Ernakulam District”, IOSR Journal of Nursing and Health Science, 15(2), pp. 54–58. doi:10.9790/1959-1502025458.

The Effect of Stretching Exercise on Reducing the Intensity of Osteoarthritis Knee Joint Pain in the Elderly in Kutambaru Village, Munthe District, Karo Regency (2025), Contemporary Journal of Applied Sciences, 3(8). doi:10.55927/cjas.v3i8.65.

Technical documentation

Crossref: crossref.org/documentation/retrieve-metadata/rest-api · OpenAlex: developers.openalex.org/api-reference/introduction · Hugging Face: huggingface.co/docs/inference-providers/en/index · LangGraph: docs.langchain.com/oss/python/langgraph/overview

Run evidence: user-provided live research export, Run ID 0e82cee494a64489bb3425c0b550d7b4 (11 October 2026).